

最终版本_POS_垂直口试复习笔记

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用途：按老师真实口试风格重组。老师通常不是按 Level 1/2/3 横着问，而是选一个大领域后一直往下追问，直到你不会，再换领域。

背诵方法：

- 每条链先背 **Opening answer**，这是保底答案。
- 老师继续追问时，再按 **Vertical follow-up** 往下答。
- 标注 **L3 / optional** 的内容，如果时间不够，只记关键词，不强背推导。
- 公式必须能写，参数必须能解释。

Sources are from the teacher PPT-based note:

- POS_oral_exam_review_from_PPT.md
- Main PPT sources: Pos01_Intro , Pos03_RF , Pos05-11_GNSS , Pos12_Inertial , Pos13-14_Indoor , Pos15_LBS , PosApp1_LS , PosApp2_Kalman

Chain 1. General Positioning Concepts

1 Define accuracy, kinematic/static positioning, real-time/post-processing, and outdoor/indoor positioning.

Opening Answer

Accuracy is the quality of positioning, defined as the error of the estimated position with respect to the true position. It is usually described for horizontal and vertical components and associated with a confidence level.

Kinematic positioning means that the receiver is moving, and positions are estimated epoch by epoch, typically at 1 Hz.

Static positioning means that the receiver is stationary during the observation session, and the position is estimated from observations over time.

Real-time positioning means that data processing is performed in real time, typically for navigation.

Post-processing means that data are acquired and stored first, and processed later, typically for surveying or monitoring.

Outdoor positioning is usually in open sky, where GNSS can satisfy many accuracy and latency requirements.

Indoor positioning is inside buildings, where GNSS is not available or is strongly corrupted, so other technologies are needed.

Vertical Follow-Up

1-1 Distance Formula

$$d(P_1, P_2) = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2 + (Z_2 - Z_1)^2}$$

If a signal travel time is (T):

$$d = cT$$

Parameters:

- (d): distance.
- (c): propagation speed of electromagnetic signal.
- (T): signal travel time.

1-2 User Requirements

Mention:

- accuracy;
- latency;
- update frequency;
- availability;
- coverage;
- cost;
- user needs.

Chain 2. Reference System, Reference Frame, ITRS

Define reference system and reference frame. What is ITRS?

Opening Answer

It is divided for the **global rs** and **local rs**,

global reference system cover the earth and are realized by the global permanent station's network. They are used for global georeferencing, Earth geodynamics, satellite orbits and

station positions

Local reference systems are defined at continental or national or local scale, are used for surveying, mapping, monitoring, and positioning.

A reference frame is the practical realization of the system, it defines the coordinates of the stations at the reference network.

ITRS is the International Terrestrial Reference System. - It is geocentric, Earth-fixed right-handed Cartesian reference system, its unit of length is the meter, and it is consistent with the TCG scale. Its realization is the ITRF.

2-1 Permanent Stations, Time Modelling of Coordinates

Permanent station coordinates are not just fixed numbers; they are modeled in time using position at a reference epoch and velocity.

永久测站坐标并非单纯的固定数值，而是借助参考历元时刻的位置与速度，从时间维度上进行建模得到的。

For permanent stations:

$$\mathbf{x}(t) = \mathbf{x}(t_0) + \dot{\mathbf{x}}(t - t_0)$$

Parameters:

- $\mathbf{x}(t)$: coordinates at epoch t .
 - $\mathbf{x}(t_0)$: coordinates at reference epoch.
 - $\dot{\mathbf{x}}$: station velocity. 站点速度
 - $t - t_0$: time interval.
-

Chain 3. Geodetic Coordinates, Ellipsoid, Geoid, Heights

Define geodetic coordinates in a 3D reference frame.

Opening Answer

Given a reference frame and the related ellipsoid, the geodetic coordinates of point P are (φ, λ, h) :

- φ , geodetic latitude: angle between the normal to the ellipsoid passing through P and the equatorial plane.
- λ , geodetic longitude: angle between the meridian plane passing through P and the origin meridian plane.
- h , geodetic height: distance between P and the ellipsoid surface. [Pos03 p56]

3-1 Reference Ellipsoid

A reference ellipsoid is a **smooth mathematical model** used to approximate the Earth's shape. It is used as the reference surface for geodetic coordinates, because the geoid is irregular and cannot be expressed by a simple analytical equation.

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{b^2} = 1$$

Parameters:

- (a): major semi-axis.
- (b): minor semi-axis.

Other parameters:

$$f = \frac{a - b}{a}, \quad e^2 = \frac{a^2 - b^2}{a^2}$$

3-2 Geodetic to Global Cartesian

$$X_P = (R_N + h_P) \cos \varphi_P \cos \lambda_P$$

$$Y_P = (R_N + h_P) \cos \varphi_P \sin \lambda_P$$

$$Z_P = [R_N(1 - e^2) + h_P] \sin \varphi_P$$

$$R_N = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi_P}}$$

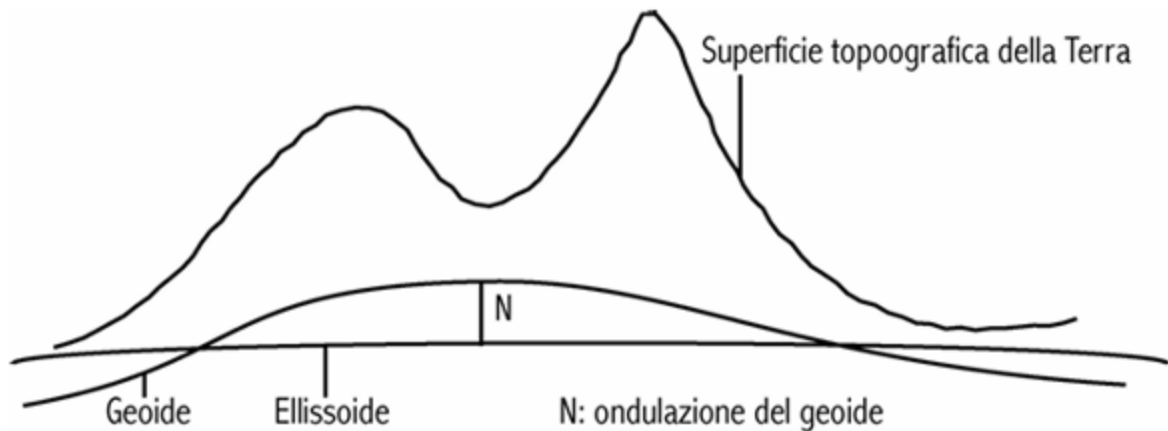
Parameters:

- X_P, Y_P, Z_P : global Cartesian coordinates of P .
- $\varphi_P, \lambda_P, h_P$: geodetic latitude, longitude, geodetic height.
- R_N : grand normal radius
- a : ellipsoid major semi-axis.
- e : ellipsoid eccentricity /ɪˈlɪpsɔɪd ˌɛksənˈtrɪsəti/. [Pos03 p57]

3-3 Geoid and Height Transformation

Geoid: The geoid is an equipotential surface of the Earth gravity field. It is close to the mean-sea-level. It is a physical surface, not a simple mathematical surface, because the geoid is irregular and cannot be modeled with simple analytical functions.

Geoid undulation N: the height of the geoid with respect to the reference ellipsoid.



GNSS gives ellipsoidal/geodetic height (h). Engineering applications need orthometric height (H).

$$h = H + N$$

$$H = h - N$$

Say this carefully:

- (h): geodetic- ellipsoidal height.
- (H): orthometric height.
- (N): geoid undulation.

3-4 Equator vs Equatorial Plane

Equatorial plane: plane used as reference for geodetic latitude.

The equator is the intersection of the equatorial plane with the ellipsoid or Earth surface.

Chain 4. Local Cartesian, Body Frame, Local Level

Explain Local Cartesian, Body Frame and Local Level reference systems.

Opening Answer

The Local Cartesian reference system is a local 3D frame, the coordinates are East, North, Up coordinates. Given a local origin point for the interest area, nearby points are expressed as East, North and Up with respect to this origin.

The body frame is a local 3D reference system attached to the vehicle, so it moves and rotates together with the body. A typical definition:

中文：车身框架用于导航。一个典型的定义：

- Origin in a generic point inside the body, for example the barycenter.
- X : main vehicle axis, motion direction.
- Y : perpendicular to X in the vehicle plane.
- Z : perpendicular to the vehicle plane. [Pos03 p66]

Local Level is typical for surveying. Its (Z) axis is oriented along the gravity vertical, and the (X - Y) plane is horizontal.

LC 是 East-North-Up, 跟地点走；BF 固定在载体/车辆上, 跟车走；LL 用重力垂线定义, 典型用于测量。

Vertical Follow-Up

4-1 Body Frame to Local Cartesian

Use roll, pitch and yaw:

- roll α : rotation around X .
 - 滚动 α : 围绕 X 旋转。
- pitch β : rotation around Y .
 - 俯仰 β : 围绕 Y 旋转。
- yaw γ : rotation around Z . [Pos03 p67]
 - 偏航 γ : 围绕 Z 旋转。 [Pos03 p67]

$$\mathbf{x}_{LC} = R_Z(\gamma)R_Y(\beta)R_X(\alpha)\mathbf{x}_{BF}$$

4-2 Local Level to Local Cartesian

- It needs vertical deflection components (ξ, η) and a horizontal angle α to transform to LC. [Pos03 p69–p70]
- 中文：它需要垂直偏转分量 (ξ, η) 和水平角 α 才能转换为LC。 [Pos03 p69–p70]

Memory:

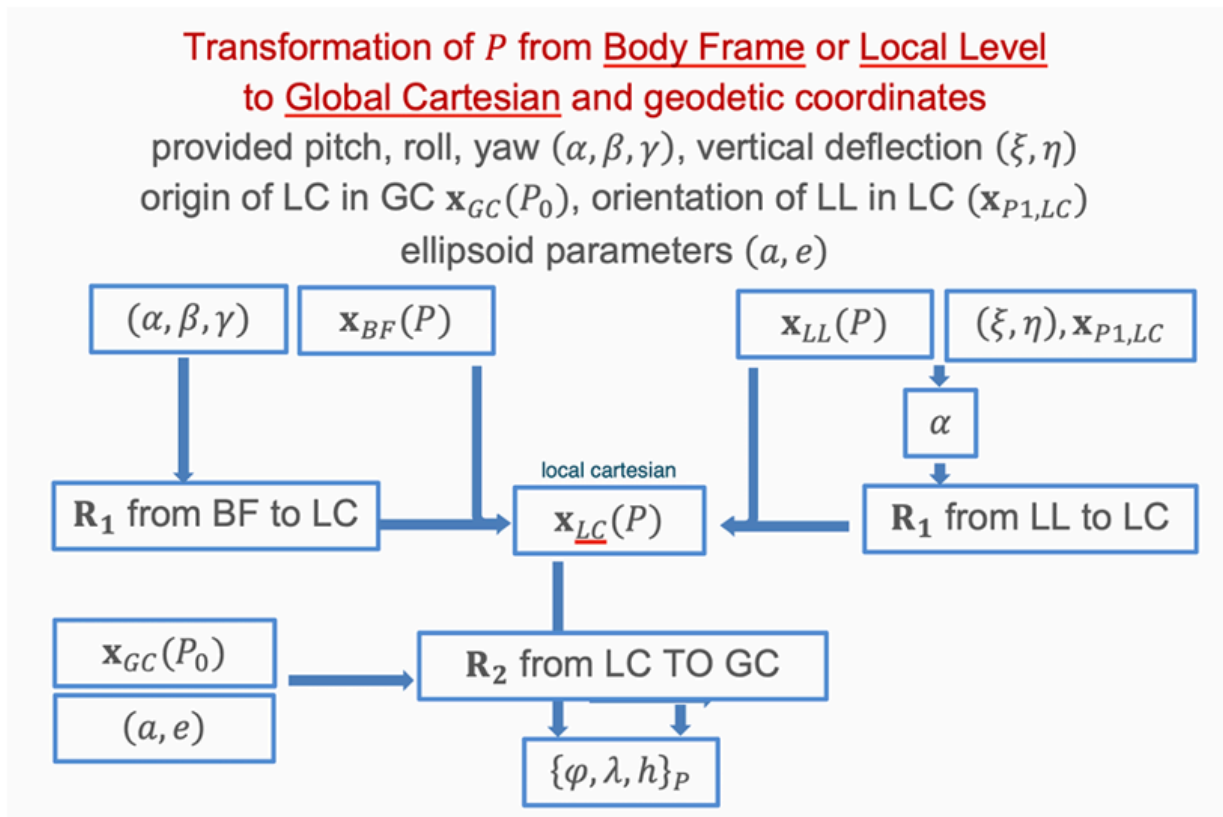
- LC uses ellipsoid normal.
- LL uses gravity vertical.

4-3 Full Transformation Chain

$$BF/LL \rightarrow LC \rightarrow GC \rightarrow (\varphi, \lambda, h)$$

Inputs:

- roll, pitch, yaw for BF;
- vertical deflection for LL;
- local origin in GC;
- ellipsoid parameters (a, e).



4-4 Covariance Propagation

For a rotation:

$$C_{new} = R C_{old} R^T$$

Parameters:

- C_{old} : covariance matrix in original frame.
- B : rotation matrix from old frame to new frame.
- C_{new} : covariance matrix in new frame.

Chain 5. GNSS Constellations, Signals, Codes and Carriers

Give a general description of GPS, GLONASS, BeiDou and Galileo, and GNSS signals.

Opening Answer

GNSS provides navigation data and observables for users.

The main observables are pseudoranges and phase observations.

The main constellations are GPS from the USA, GLONASS from Russia, Galileo from Europe, and BeiDou from China. In the past positioning was mainly based on GPS; now positioning uses GPS, Galileo, BeiDou and GLONASS together.

GNSS signals contain binary codes, sinusoidal carrier phases, and navigation messages.

Binary codes are sequences of +1 and -1 pulses and are used for pseudorange observations.

Carrier phases are sinusoidal signals and are used for high precision positioning.

Vertical Follow-Up

5-1 Differences Between Constellations

Mention only PPT-safe differences:

- ownership: USA, Russia, Europe, China;
- GPS uses CDMA- code division multiple access; GLONASS historically uses FDMA - frequency division multiple access;
- ionospheric models: GPS Klobuchar, GLONASS no ionospheric model, BeiDou BDGIM, Galileo NeQuick.

5-2 Binary Codes

A binary code is a sequence of pulses equal to +1 and -1. The receiver compares the received code with local replica to obtain the code observation or pseudorange.

5-3 Carrier Phases

Carrier phase observations is ppp's observation, it have millimeter-level electronic noise, but they include integer ambiguities.

Chain 6. SPP Vertical Chain

Describe Standard Point Positioning: observations, unknowns and biases.

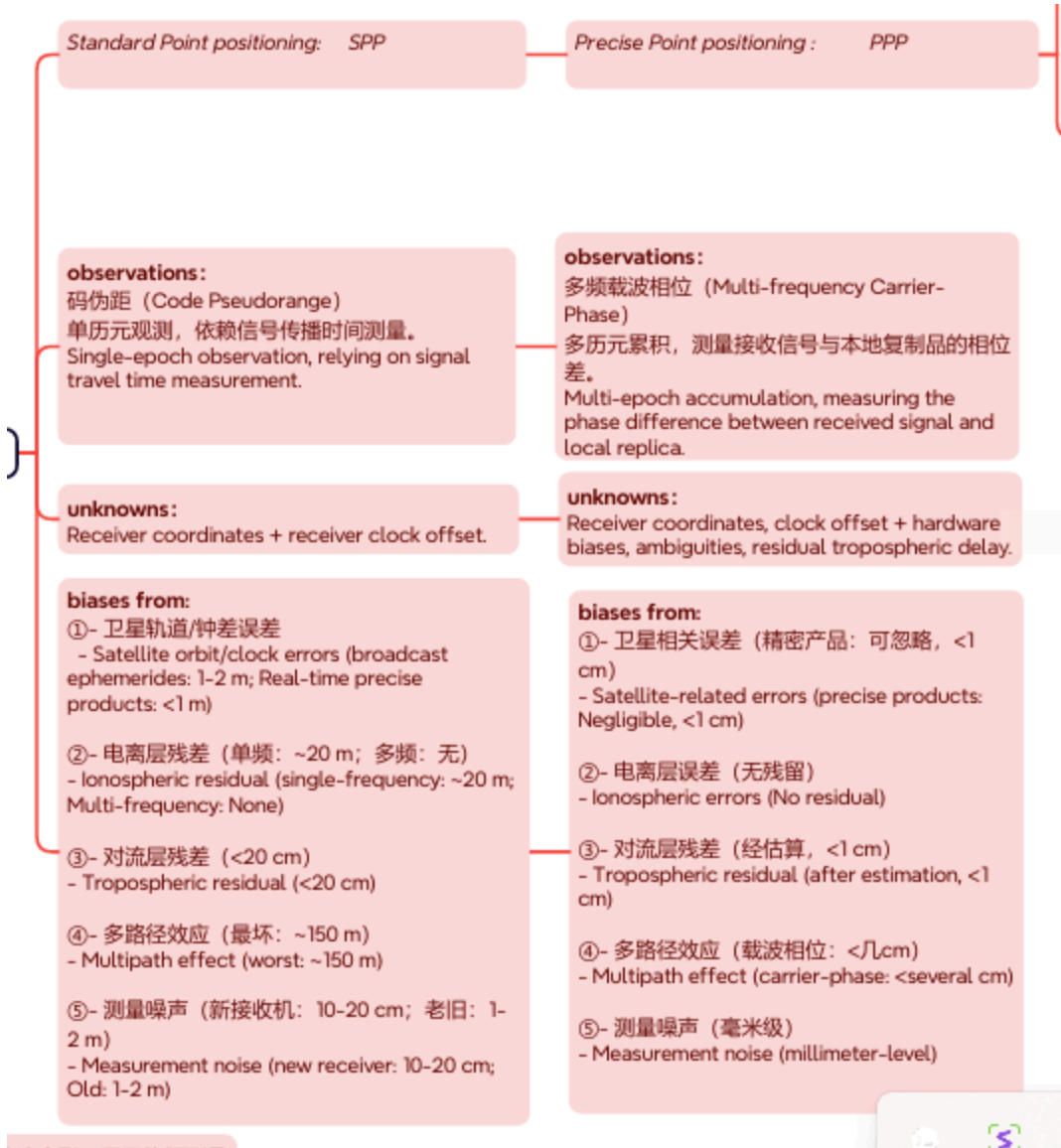
Opening Answer

Standard Point Positioning, or SPP, is based on code observations, also called pseudoranges.

SPP is single epoch, single constellation and single frequency.

With m satellites in view, there are m code observations.

The unknowns are the receiver coordinates (X_R, Y_R, Z_R) and the receiver clock offset (dt_R). Therefore, at least 4 satellites are needed.



6-1 Vertical Follow-Up 1: Pseudorange Equation

$$P_{R,i}^{S,G}(t) = \rho_R^S(t) + c[dt_R(t) + d_{R,i}^G - dt^S(t) - d_i^{S,G}] + T_R^S(t) + I_{R,i}^S(t) + \varepsilon$$

Parameters:

- $P_{R,i}^{S,G}$: code/pseudorange observation from receiver R to satellite S , constellation G , frequency i .
 - ρ_R^S : geometric distance between receiver and satellite.
- c : speed of light.
- dt_R : receiver clock offset.
- $d_{R,i}^G$: receiver hardware bias for constellation/frequency.
- dt^S : satellite clock offset.

- $d_i^{S,G}$: satellite hardware bias.
- T_R^S : tropospheric delay.
- $I_{R,i}^S$: ionospheric delay.
- ε : residual modeling error and measurement noise. [Pos09 p2]

6-2 Vertical Follow-Up 2: SPP Least Squares

SPP equations are nonlinear with respect to receiver coordinates, so linearization is needed around approximate coordinates (X_R^0, Y_R^0, Z_R^0).

Unknown vector:

$$\Delta x = \begin{bmatrix} \Delta X_R \\ \Delta Y_R \\ \Delta Z_R \\ c dt_R \end{bmatrix}$$

Linearized equation:

$$\Delta P_R^S = e_x^S \Delta X_R + e_y^S \Delta Y_R + e_z^S \Delta Z_R + c dt_R + \varepsilon$$

Matrix form:

$$\Delta \mathbf{P} = A \Delta x + \varepsilon$$

Least Squares solution:

$$\hat{\Delta x} = (A^T Q^{-1} A)^{-1} A^T Q^{-1} \Delta \mathbf{P}$$

Parameters:

- (A): design matrix; each row contains line-of-sight components and one clock column.
- (Q): observation cofactor/weight matrix.

Chain 7. Atmospheric Effects: Ionosphere, Troposphere, Multipath

Introduce atmospheric effects in GNSS.

Opening Answer

The main atmospheric effects in GNSS are ionospheric delay, tropospheric delay and multipath.

The ionosphere is dispersive and frequency dependent. The troposphere is due to air and water vapor in the neutral atmosphere.

Multipath is caused by reflected GNSS signals near the receiver antenna.

7-1 Vertical Follow-Up 1: Ionospheric Effect

The ionospheric effect is due to free electrons between about 100 and 1000 km altitude.

It is frequency dependent:

$$I_R^S = \pm \frac{A}{f^2} TEC_R^S$$

Parameters:

- (I_R^S): ionospheric effect between receiver and satellite.
- (A): constant.
- (f): carrier frequency.
- (TEC_R^S): Total Electron Content along the signal path.
- plus sign: code observations.
- minus sign: carrier phase observations.

7-2 Vertical Follow-Up 2: Ionosphere-Free Combination

Since ionospheric delay depends on two frequencies can be linearly combined to remove the first-order ionospheric delay.

For code observations:

$$P_3 = \alpha_1 P_1 + \alpha_2 P_2$$

$$\alpha_1 = \frac{f_1^2}{f_1^2 - f_2^2}$$

$$\alpha_2 = -\frac{f_2^2}{f_1^2 - f_2^2}$$

Therefore:

$$P_3 = \frac{f_1^2}{f_1^2 - f_2^2} P_1 - \frac{f_2^2}{f_1^2 - f_2^2} P_2$$

In this combined observation, the first-order ionospheric delay is cancelled.

7-3 Vertical Follow-Up 4: Residual Troposphere in PPP

In PPP:

$$T_R^S(t) = T_{0,R}^S(t) + \delta T_R^S(t)$$

Residual model:

$$\delta T_R^S(t) = \delta T_Z(t) m(el_R^S)$$

parameters:

- $T_R^S(t)$: total tropospheric delay.
- $T_{0,R}^S(t)$: standard model value.
- $\delta T_R^S(t)$: residual model error along satellite direction.
- $\delta T_Z(t)$: zenith tropospheric residual, estimated parameter.
 - 中文: $\delta T_Z(t)$: 天顶对流层残差, 估计参数。
- $m(el_R^S)$: mapping function depending on satellite elevation.
 - 中文: $m(el_R^S)$: 取决于卫星高程的映射函数。
- el_R^S : satellite elevation. [Pos10 p16-p17]
 - 中文: el_R^S : 卫星高程。 [Pos10 p16-p17]

The zenith tropospheric residual is estimated as an unknown together with coordinates, clock offsets and ambiguities.

7-4 Vertical Follow-Up 5: Multipath

Multipath is typical in urban. It is due to reflection of the GNSS signal when approaching the receiver antenna.

The receiver may receive direct and reflected signals, disturbing the observation. Its mathematical modelling is complicated and not intuitive.

- some centimeters for phase observations;
- several meters for code observations;
- periodical behavior, period about 20 minutes.

Chain 8. Phase Observations, Ambiguity, PPP

Explain carrier phase observations and integer ambiguity.

Opening Answer

In carrier phase observations, the receiver can observe the fractional part of the incoming carrier phase, but cannot observe the integer number of full cycles between satellite and receiver. Therefore, the phase observation is affected by an unknown integer ambiguity.

If there is no loss of lock, the receiver can count phase changes from epoch to epoch, so the initial ambiguity is unknown but constant in time. With enough epochs, coordinates and ambiguities can be estimated.

If signal lock is lost, a new ambiguity appears. This is equivalent to a cycle slip.

载波相位只能观测小数周，不能知道卫星到接收机之间完整周数，所以有整数模糊度。不中断锁定时，初始模糊度未知但随时间常数；失锁后出现新的模糊度，即 cycle slip。

8-1 Vertical Follow-Up 1: Phase Observation Equation

$$\Phi_{R,i}^{S,G}(t) = \rho_R^S(t) + c[dt_R(t) + b_{R,i}^G - dt^S(t) - b_i^{S,G}] + T_R^S(t) - I_{R,i}^S(t) + \lambda_i N_{R,i}^{S,G} + \varepsilon_\Phi$$

Parameters:

- $\Phi_{R,i}^{S,G}$: carrier phase observation in meters.
- ρ_R^S : geometric range.
- c : speed of light.
- dt_R, dt^S : receiver and satellite clock offsets.
- $b_{R,i}^G, b_i^{S,G}$: phase hardware biases.
- T_R^S : tropospheric delay.
- $I_{R,i}^S$: ionospheric delay, opposite sign with respect to codes.
- λ_i : carrier wavelength.
 - 中文: λ_i : 载波波长。
- $N_{R,i}^{S,G}$: integer ambiguity.
 - 中文: $N_{R,i}^{S,G}$: 整数歧义。
- ε_Φ : phase noise, about 1-2 mm. [Pos10 p5-p9]

8-2 Vertical Follow-Up 2: PPP

PPP is based on multi-frequency phase observations, it is multi-epoch accumulation, measuring the phase difference between received signal and local signal.

ppp - unknowns are:

1. 3 coordinates;
2. receiver clock offsets
3. constellation hardware biases;
4. ambiguities;
5. at least one zenith tropospheric residual.



8-3 Vertical Follow-Up 3: Ambiguity Fixing

The receiver can observe the fractional part of the incoming carrier, but not the integer number of cycles from satellite emission to receiver epoch. This unknown integer is the ambiguity.

Ambiguity fixing means estimating carrier phase ambiguities and fixing them to integer values. It is essential for high precision phase processing and especially for phase double differences.

Chain 9. Relative Positioning and Double Differences

Explain single and double differences in relative GNSS positioning.

Opening Answer

GNSS relative positioning primarily uses **carrier phase measurements** to achieve high-precision positioning,

A single difference is formed between two receivers observing the same satellite, or between two satellites for the same receiver. It reduces common errors such as satellite or receiver clock terms.

A double difference combines two receivers and two satellites. It removes receiver and satellite clock errors and reduces common atmospheric and orbit errors,

Chain 10. GNSS Orbits and Ephemerides

Explain Keplerian orbits and broadcast ephemerides.

Opening Answer

A Keplerian orbit is stationary and defined by 6 parameters.

The usual six parameters:

- a : semi-major axis.
- e : eccentricity.
- i : inclination.
 - 中文: i : 倾斜度。
- Ω : right ascension of ascending node.
 - 中文: Ω : 升交点赤经。
- ω : argument of perigee.
 - 中文: ω : 近地点角。
- M_0 or anomaly at reference epoch.
 - 中文: M_0 或参考时期的异常。

Actual GNSS orbits are not perfectly Keplerian because of perturbations.

Perturbations require corrections and periodic updates of broadcast ephemerides.

Satellite ephemerides are parameters that allow computation of satellite positions. Broadcast ephemerides are transmitted in the navigation message and allow real-time positioning.

Chain 11. INS Components, Sensors, Errors

Describe an Inertial Navigation System and its main components.

Opening Answer

An Inertial Navigation System consists of 3 accelerometers, 3 gyroscopes, hardware for collecting data, and software for real-time processing.

The IMU output contains 6 quantities: 3 linear accelerations and 3 angular rates.

Accelerometers measure the forces acting on a proof mass, or the specific acceleration along three orthogonal axes.

Gyroscopes measure the angular rate of sensor rotation with respect to an inertial reference system.

11-1 Vertical Follow-Up 1: Accelerometers

Accelerometers measure the forces acting on a proof mass.

Open-loop accelerometers measure the displacement of the proof mass caused by external forces.

Closed-loop accelerometers keep the proof mass in equilibrium (in balanced) by generating a force opposite to the applied force.

If formula is asked:

$$a = \frac{F}{m}$$

Parameters:

- (F): force along sensor axis.
- (m): proof mass.
- (a): acceleration.

11-2 Vertical Follow-Up 2: Gyroscopes

Gyroscopes measure angular rate of sensor rotation with respect to an inertial reference system.

Optical gyroscopes are based on the Sagnac 萨尼亚克 effect.

[Two light beams travel in opposite directions in a rotating fiber coil; [Pos12 p13]]

formula:

$$\Delta\Phi = \frac{4\Omega A}{\lambda c}$$

Parameters:

- $\Delta\Phi$: phase shift between opposite light beams.
 - 中文: $\Delta\Phi$: 相对光束之间的相移。

- Ω : angular rate.
 - 中文: Ω : 角速率。
- A : area enclosed by the optical path.
 - 中文: A : 光路包围的区域。
- λ : wavelength.
 - 中文: λ : 波长。
- c : speed of light.
 - 中文: c : 光速。

11-3 Vertical Follow-Up 3: Sensor Error Models

Accelerometer error:

$$\delta a = b + \lambda a + c_T \Delta T + v$$

Gyroscope error:

$$\delta \omega = b + \lambda \omega + c_T \Delta T + v$$

Parameters:

- b : bias.
- λ : scale factor.
- c_T : constant depending on temperature.
 - 中文: c_T : 取决于温度的热常数。
- ΔT : temperature variation.
- v : measurement noise. [Pos12 p11, p14]
 - 中文: v : 测量噪声。 [Pos12 p11, p14]

Chain 12. Indoor Positioning

Describe indoor positioning applications and user requirements.

Opening Answer

Indoor positioning is positioning inside buildings, where GNSS is not available or signals are severely corrupted. It needs other infrastructure and technologies.

12-1 Vertical Follow-Up 1: Known Nodes and Sensors

Known nodes are fixed points whose coordinates are known, where markers or sensors are installed.

Unknown nodes are mobile users with sensors whose coordinates have to be estimated.

An active/transmitter sensor transmits signals.

A passive/receiver sensor receives or reads signals.

12-2 Vertical Follow-Up 2: RSSI

RSSI means Received Signal Strength Indicator. It is the received signal strength averaged over a certain period, usually expressed in decibels.

$$dB = 10 \log_{10} \left(\frac{P}{P_0} \right)$$

Parameters:

- P : measured power.
- P_0 : reference power. [Pos14 p2]

Physical attenuation model物理衰减模型:

THE physical attenuation is the basic formula, used to describing that the strength of signals attenuates as the propagation distance increases.

$$P_R = k P_T \frac{G_T G_R}{4\pi d_{TR}^p}$$

Parameters:

- P_R, P_T : received and transmitted signal strengths.
- G_T, G_R : transmitter and receiver antenna gains.
- k : system correction constant
- d_{TR} : distance between transmitter and receiver.
- p : path loss exponent.
 - p : 路径损耗指数。
- $p = 2$ in open space; indoor typically $4 \leq p \leq 6$, but in narrow corridors $p < 2$ because multipath cumulates. [Pos14 p3]
 - 中文: $p = 2$ 在开放空间; 室内通常为 $4 \leq p \leq 6$, 但在狭窄的走廊 $p < 2$, 因为多径累积。 [Pos14 p3]

12-3 Vertical Follow-Up 3: ToA

Time of Arrival is based on travel time from transmitter to receiver, like GNSS pseudoranges. The main technical issue is clock synchronization.

$$ToA_i^u = \tau_i^u + \delta_u - \delta_i$$

Parameters:

- ToA_i^u : measured time of arrival between transmitter i and user u .
- τ_i^u : signal travel time.
- δ_u : receiver/user clock offset.
- δ_i : transmitter clock offset.

If transmitters are synchronized known nodes:

$$ToA_i^u = \tau_i^u + \delta_u$$

Unknowns: user coordinates plus one receiver clock offset.

Biases: multipath, NLoS, signal slowing through walls and obstacles.

12-4 Vertical Follow-Up 4: Cell of Origin and Centroid

Cell of Origin: the mobile is located at the most probable known node, for example highest RSSI or smallest ToA.

Centroid:

$$\mathbf{x}_u = \sum_{i=1}^N w_i \mathbf{x}_i, \quad \sum_{i=1}^N w_i = 1$$

For RSSI, weights are proportional to observations. For ToA, weights are inversely proportional to observations.

12-5 Vertical Follow-Up 5: Fingerprinting

Fingerprinting has two steps:

1. calibration;
2. usage. [Pos14 p14]

Calibration:

- choose M control points in the area;
 - 中文：选择区域内的 M 控制点；
- estimate their coordinates accurately;
 - 中文：准确估计它们的坐标；
- in each control point store signals from/to all N known nodes.
 - 中文：在每个控制点中存储来自/到所有 N 已知节点的信号。

Stored data for each control point:

$$\mathbf{x}_{CP_i} = [x_{CP_i}, y_{CP_i}]^T, \quad \mathbf{r}_{CP_i} = [r_{KN1}, r_{KN2}, \dots, r_{KNN}]^T$$

Usage:

- at unknown position, the mobile measures a signal vector $\mathbf{s}(t_u)$;
 - 中文：在未知位置，移动设备测量信号矢量 $\mathbf{s}(t_u)$;
- compare it with all stored fingerprints;
 - 中文：与所有存储的指纹进行比较;
- assign the position of the control point minimizing the difference:
 - 中文：分配控制点的位置以最小化差异:

$$\mathbf{x}_u = \mathbf{x}_{CP_i} \quad \text{such that} \quad \|\mathbf{s}(t_u) - \mathbf{r}_{CP_i}\| = \min$$

Chain 13. LBS and Path Optimization

Give a general description of Location Based Services.

Opening Answer

Location Based Services are fully location-aware applications delivering geolocated services through client-server or peer-to-peer architectures by wireless and satellite communication networks.

They provide geographically oriented data and information services to users across mobile telecommunication networks.

13-1 Vertical Follow-Up 1: User Needs

Users determine what information should be provided and how systems should be designed.

Questions:

- Who are the users?
- What are their needs?
- When and where do they need it?

Examples:

- tourists search shops/restaurants/galleries;
- impaired people need accessible paths;
- rescue users need rescuers to know their position quickly.

13-2 Vertical Follow-Up 2: Data Modelling

Geographic model:

- points, lines, areas, volumes;
- georeferenced in a reference system;
- with attributes.

Symbolic model:

- locations are classes with attributes;
- objects are members of locations;
- relationships are used for queries such as optimal paths.

13-3 Vertical Follow-Up 3: Road Topology

Roads are sequences of connections between intersections. Intersections and terminal points are nodes.

Stored data:

- regulatory/physical/geometrical attributes;
- connections;
- rights: allowed yes/no;
- costs: length, fee, time.

Important rules:

- if road conditions change, a road is split into different connections;
- bridges/tunnels can overlap geographically without topological intersection;
- nodes store entering connections and turn rules/costs.

13-4 Vertical Follow-Up 4: Dijkstra Algorithm

Dijkstra algorithm finds the minimum-cost path from origin node to destination node.

Definitions:

- estimated node: cost from origin has been estimated at least once;
- explored node: already explored;
- current explored node: node expanded in current iteration.

Algorithm:

1. Start from origin as current explored node.
2. Estimate costs of connected nodes.

3. Choose the nearest estimated but unexplored node.
4. Update connected nodes with minimum between old cost and new cost.
5. Store minimum path and explored node list.
6. Stop when destination becomes current explored node.

Stored data:

- node;
 - explored yes/no;
 - estimated yes/no;
 - minimum cost;
 - path.
-

Chain 14. Kalman Filtering Optional Chain

Explain the model, observations and estimation principle of Kalman filtering.

Opening Answer

The purpose of Kalman filtering is to predict the evolution of a dynamic system in real time and update the previous estimate using new observations and the dynamic model.

Dynamic model:

$$\mathbf{x}_{t+1} = T_{t+1}\mathbf{x}_t + \xi_{t+1}$$

Observation model:

$$\mathbf{y}_{t+1} = A_{t+1}\mathbf{x}_{t+1} + \nu_{t+1}$$

Prediction:

$$\mathbf{x}_{t+1|t} = T_{t+1}\mathbf{x}_{t|t}$$

Innovation:

$$\eta_{t+1} = \mathbf{y}_{t+1} - A_{t+1}\mathbf{x}_{t+1|t}$$

Update:

$$\mathbf{x}_{t+1|t+1} = \mathbf{x}_{t+1|t} + G_{t+1}\eta_{t+1}$$

Fast Vertical Exam Routes

Route A: RF / Coordinates

1. System vs Frame.
2. ITRS / ITRF.
3. Reference ellipsoid.
4. Geodetic coordinates (φ, λ, h).
5. Cartesian-geodetic formula.
6. Geoid undulation and height conversion.
7. LC / BF / LL transformation.
8. Covariance propagation.

Route B: GNSS / SPP

1. GNSS constellations and signals.
2. Binary codes and pseudoranges.
3. SPP observations/unknowns/biases.
4. Pseudorange equation.
5. Least Squares linearization.
6. SPP error budget.

Route C: GNSS / Atmosphere / PPP

1. Ionosphere: free electrons, frequency dependent.
2. Ionospheric formula ($I = \pm A \text{ TEC} / f^2$).
3. Ionosphere-free combination.
4. Troposphere: dry/wet, ZTD, residual.
5. PPP residual troposphere estimation.
6. Multipath.
7. Phase ambiguity.
8. PPP unknowns and why multi-epoch.

Route D: INS

1. INS components.
2. IMU output.
3. Accelerometer principle.
4. Gyroscope principle.
5. Error models.
6. Navigation equations only if asked.

Route E: Indoor / LBS

1. Indoor applications and requirements.
 2. Known/unknown nodes, active/passive sensors.
 3. RSSI formula.
 4. ToA formula.
 5. CoO / Centroid.
 6. Fingerprinting.
 7. Wi-Fi / RFID.
 8. LBS definition.
 9. Road topology.
 10. Dijkstra.
-

Last-Day Priority

Must know:

- Geodetic coordinates and height transformation.
- SPP observations, unknowns, biases, minimum 4 satellites.
- Pseudorange equation.
- Ionospheric formula and ionosphere-free combination.
- Tropospheric effect and ZTD.
- Multipath.
- Phase ambiguity and PPP unknowns.
- INS components, accelerometer, gyroscope.
- Indoor known nodes, RSSI, ToA, fingerprinting.
- LBS and Dijkstra.

Can skip if no time:

- Full INS navigation equations.
- Full Kalman derivation.
- Detailed Helmert / covariance derivation.
- Full DD batch processing derivation.